



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 13: Producer Theory III

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ICS2523 Microeconomics

Lecture 13

Last lecture we studied

- Elasticity of Substitution
- Returns to scale
- Definitions of costs
- Isocosts curves
- Minimization of costs

In today's lecture, we will learn more about
Costs and Profit Maximization

- Firm's expansion path
- Cost functions
- Economies of Scale
- Distinguish the Short- and Long-run

Homework 1 L12: solution

If the production function is given by a generalized Leontief production function:

$$q = f(k, l) = k + l + 2\sqrt{kl}$$

- a. Graph the isoquants
- b. Calculate the Elasticity of substitution
- c. Analyse the returns to scale

Homework 1 L12 : solution

a. Graph the isoquants

$$q = f(k, l) = k + l + 2\sqrt{kl}$$

$$q = (\sqrt{k} + \sqrt{l})^2$$

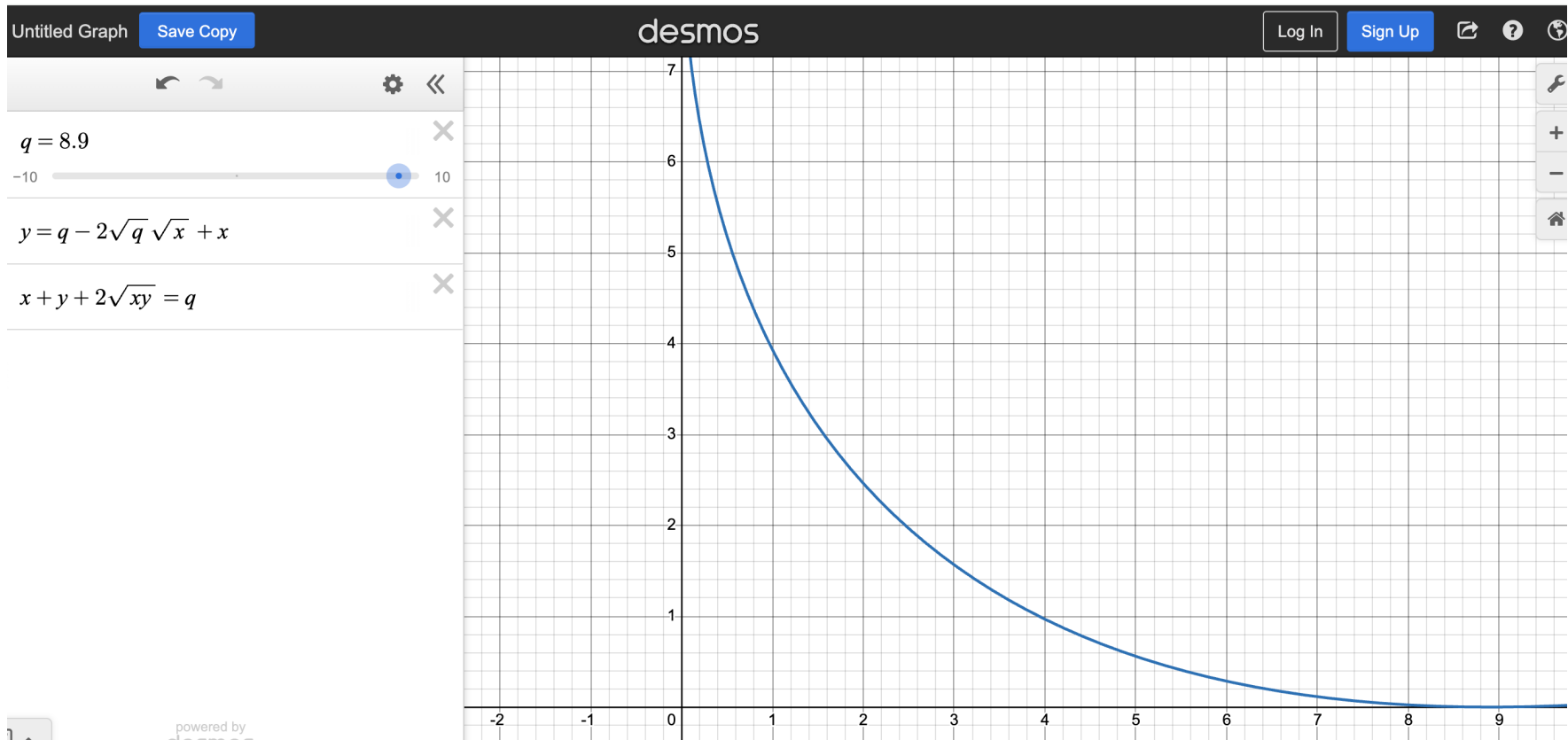
$$\sqrt{q} = \sqrt{k} + \sqrt{l}$$

$$\sqrt{q} - \sqrt{l} = \sqrt{k}$$

$$k = +l + q - 2\sqrt{lq}$$

Homework 1 L12: solution

- a. Graph the isoquants: $k = +l + q - 2\sqrt{lq}$
k per period



/ per period

Homework 1 L12: solution

c. Analyse the returns to scale

$$q = f(k, l) = k + l + 2\sqrt{kl}$$

$$f(tk, tl) = tk + tl + 2\sqrt{tktl}$$

$$f(tk, tl) = t(k + l + 2\sqrt{kl}) = tf(k, l)$$

→ Constant returns to scale

Homework 1 L12: solution

b. Calculate the Elasticity of substitution

There are two ways of calculating this value:

First way:

$$q = f(k, l) = k + l + 2\sqrt{kl}$$

$$RTS = \frac{MP_l}{MP_k} = \frac{1 + (k/l)^{1/2}}{1 + (k/l)^{-1/2}} = \frac{\sqrt{l}/\sqrt{l} + \sqrt{k}/\sqrt{l}}{\sqrt{k}/\sqrt{k} + \sqrt{l}/\sqrt{k}} = \frac{(\sqrt{l} + \sqrt{k})/\sqrt{l}}{(\sqrt{k} + \sqrt{l})/\sqrt{k}} = \frac{\sqrt{k}}{\sqrt{l}}$$

$$RTS = \frac{\sqrt{k}}{\sqrt{l}}$$

$$\ln(RTS) = \frac{1}{2} \ln(k/l)$$

$$\ln\left(\frac{k}{l}\right) = 2 \cdot \ln(RTS)$$

$$\sigma_{l,k} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = 2$$

Homework 1 L12: solution

b. Calculate the Elasticity of substitution

Second way: (using formula for constant returns to scale)

$$\sigma = \frac{f_k f_l}{f f_{kl}} = \frac{(1 + (k/l)^{-1/2})(1 + (k/l)^{1/2})}{(k + l + 2\sqrt{kl}) \cdot \frac{1}{2\sqrt{kl}}}$$
$$\sigma = \frac{(1 + \left(\frac{k}{l}\right)^{\frac{1}{2}} + \left(\frac{k}{l}\right)^{-\frac{1}{2}} + 1)}{(k + l + 2\sqrt{kl}) \cdot \frac{1}{2\sqrt{kl}}} = \frac{2(2 + \left(\frac{k}{l}\right)^{\frac{1}{2}} + (k/l)^{-1/2})}{\left(\left(\frac{k}{l}\right)^{\frac{1}{2}} + (k/l)^{-1/2} + 2\right)} = 2$$

Homework 2 L12: Solution

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

$$\text{CES: } q = f(k, l) = (k^\rho + l^\rho)^{\gamma/\rho}$$

Homework 2 L12: Solution

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

b) CES: $q = f(k, l) = (k^\rho + l^\rho)^{\gamma/\rho}$

$$\text{Min}_{\{k, l\}} C = r \cdot k + w \cdot l$$

s. to:

$$f(k, l) = q_0 \quad (\lambda)$$

$$\mathcal{L} = r \cdot k + w \cdot l + \lambda(q_0 - (k^\rho + l^\rho)^{\gamma/\rho})$$

Homework 2 L12: Solution

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

$$\mathcal{L} = r \cdot k + w \cdot l + \lambda(q_0 - (k^\rho + l^\rho)^{\gamma/\rho})$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial k} = r - \lambda \frac{\gamma}{\rho} (k^\rho + l^\rho)^{\gamma/\rho - 1} \rho k^{\rho-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial l} = w - \lambda \frac{\gamma}{\rho} (k^\rho + l^\rho)^{\gamma/\rho - 1} \rho l^{\rho-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q_0 - (k^\rho + l^\rho)^{\gamma/\rho} = 0$$

Homework 2 L12: Solution

Dividing the second equation by the first yields:

$$\frac{r}{w} = \frac{\frac{\gamma}{\rho} (k^\rho + l^\rho)^{\gamma/\rho - 1} \rho k^{\rho-1}}{\frac{\gamma}{\rho} (k^\rho + l^\rho)^{\gamma/\rho - 1} \rho l^{\rho-1}} = \frac{k^{\rho-1}}{l^{\rho-1}}$$
$$l^{\rho-1} = \frac{w}{r} k^{\rho-1}$$
$$k = \left(\frac{r}{w}\right)^{\frac{1}{\rho-1}} l$$

Replacing in the third equation:

$$q_0 - (k^\rho + l^\rho)^{\gamma/\rho} = 0$$

Homework 2 L12: Solution

$$q_0 = \left(\left(l \left(\frac{r}{w} \right)^{\frac{1}{\rho-1}} \right)^\rho + l^\rho \right)^{\gamma/\rho}$$

$$q_0 = l^\gamma \left(\left(\frac{r}{w} \right)^{\frac{\rho}{\rho-1}} + 1 \right)^{\gamma/\rho}$$

$$\frac{q_0}{\left(\left(\frac{r}{w} \right)^{\frac{\rho}{\rho-1}} + 1 \right)^{\gamma/\rho}} = l^\gamma$$

$$\frac{q_0}{\left(\left(\frac{r}{w} \right)^{\frac{\rho}{\rho-1}} + 1 \right)^{\gamma/\rho}} = l^\gamma$$

Homework 2 L12: Solution

$$l^* = \frac{q_0^{\frac{1}{\gamma}}}{\left(\left(\frac{r}{w}\right)^{\frac{\rho}{\rho-1}} + 1\right)^{1/\rho}}, k^* = \left(\frac{r}{w}\right)^{\frac{1}{\rho-1}} \frac{q_0^{\frac{1}{\gamma}}}{\left(\left(\frac{w}{r}\right)^{\frac{\rho}{\rho-1}} + 1\right)^{1/\rho}}$$

$$C = rk^* + wl^* = r\left(\frac{r}{w}\right)^{\frac{1}{\rho-1}} l^* + wl^*$$

$$C = \frac{q_0^{\frac{1}{\gamma}}}{\left(\left(\frac{r}{w}\right)^{\frac{\rho}{\rho-1}} + 1\right)^{1/\rho}} \left(r\left(\frac{r}{w}\right)^{\frac{1}{\rho-1}} + w\right)$$

$$C = \frac{q_0^{\frac{1}{\gamma}} w}{\left(\left(\frac{r}{w}\right)^{\frac{\rho}{\rho-1}} + 1\right)^{1/\rho}} \left(\left(\frac{r}{w}\right)^{\frac{\rho}{\rho-1}} + 1\right) = w q_0^{\frac{1}{\gamma}} \left(\left(\frac{r}{w}\right)^{\frac{\rho}{\rho-1}} + 1\right)^{(\rho-1)/\rho}$$

Firm's Expansion Path

Producer Theory

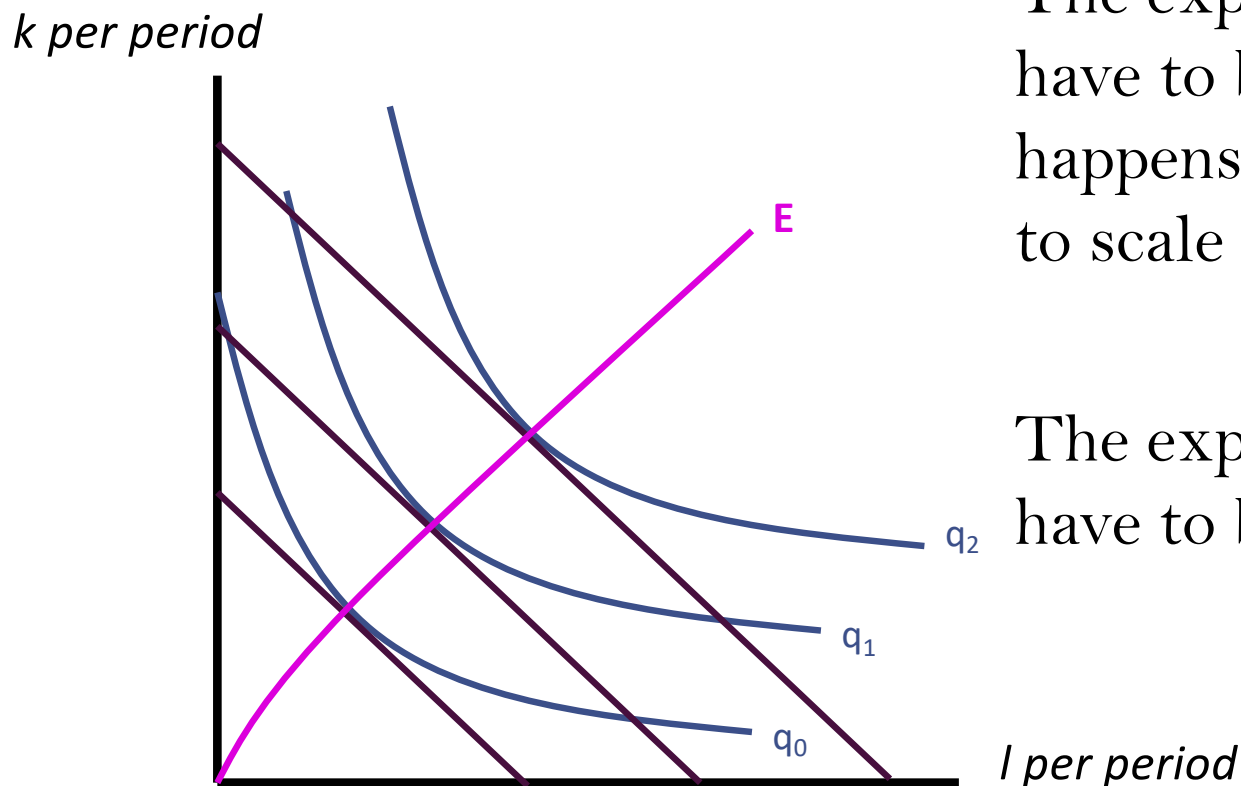
10. Firm's expansion path

The firm can determine the cost-minimizing combinations of k and l **for every level of output.**

If input costs (inputs' prices) remain constant for all amounts of k and l the firm may demand, we can trace the **locus of cost-minimizing choices** → **Firm's expansion path**

Firm's expansion path

Locus of cost-minimizing choices → Firm's expansion path



The expansion path does not have to be a straight line (it happens with a constant return to scale function, though)

The expansion path does not have to be upward-sloping.

Cost Functions

Producer Theory

11. Cost Functions

Total cost function (C): for any set of input costs and any output level, the minimum total cost incurred by the firm is

$$C = C(r, w, q)$$

As seen in the expansion path, total costs increase as the output level increases.

Average cost function (AC): total cost per unit of output

$$AC = AC(r, w, q) = C(r, w, q) / q$$

Marginal cost function (MC): cost of one more unit of output

$$MC = MC(r, w, q) = \partial C(r, w, q) / \partial q_{19}$$

Activity 1

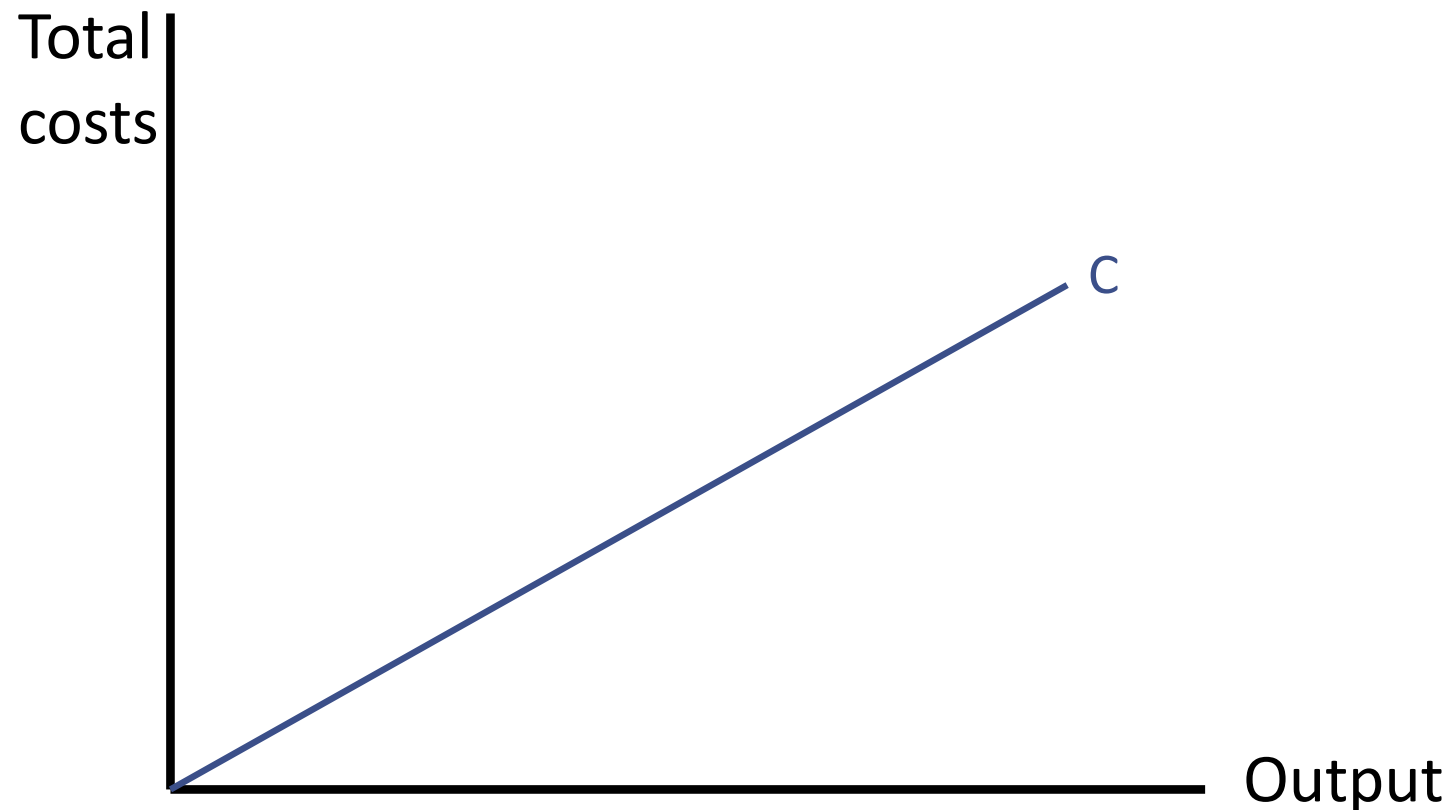
Proposition: the **average cost function** always reaches its **lowest level** at the point of production where the **marginal and average cost functions intersect**

Demonstrate this proposition using the mathematical definitions of average and marginal cost functions.

11. Cost Functions

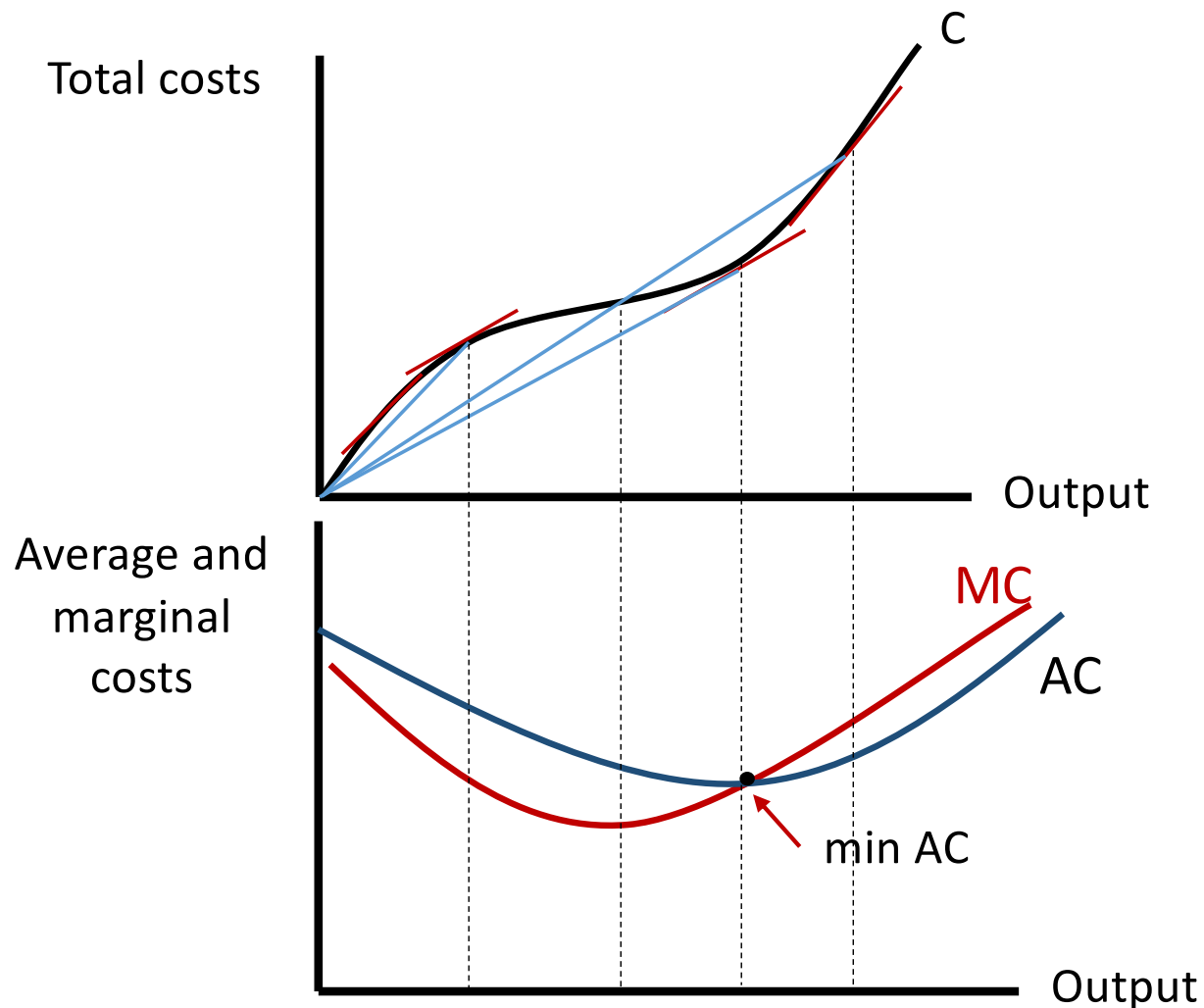
Graphical analysis of Total Costs

With **constant returns to scale**, total costs are proportional to output $\rightarrow AC = MC$ and both are constant



11. Cost Functions

With **cubic total cost curve**, total costs rise dramatically as output increases after diminishing returns set in



11. Cost Functions

Properties of cost functions

- Total cost functions are homogenous of degree 1 in the **input prices**
- Total cost functions are **non decreasing** in **q, r, and w**
- Total cost functions are **concave** in **input prices**
- AC and MC are also **homogenous degree 1** in the **input prices**

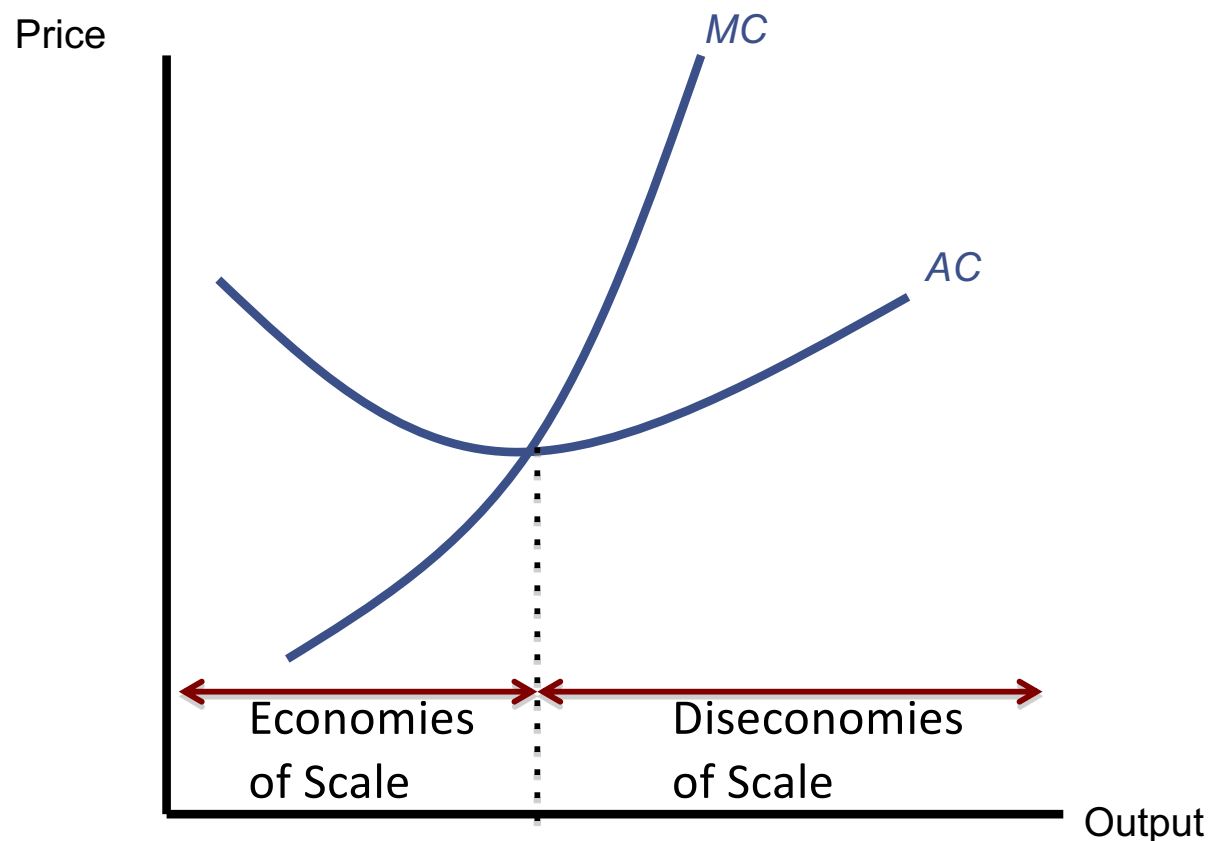
Economies of Scale

Producer Theory

12. Economies of Scale

Decreasing Average Cost ($MC < AC$) → **Economies of scale**

Increasing Average Cost ($MC > AC$) → **Diseconomies of scale**



12. Economies of Scale

Decreasing Average Cost ($MC < AC$) $\rightarrow$ Economies of scale

Increasing Average Cost ($MC > AC$) $\rightarrow$ Diseconomies of scale

Relationship with returns to scale:

- Constant returns to scale ($f(tk, tl) = tf(k, l)$)
Since the price of inputs are constant, AC is constant for different levels of production $\rightarrow$ MC and AC are horizontal lines
- Increasing returns to scale ($f(tk, tl) > tf(k, l)$)
AC is decreasing $\rightarrow$ Economies of Scale
- Decreasing returns to scale ($f(tk, tl) < tf(k, l)$)
AC should be increasing $\rightarrow$ Diseconomies of Scale

Long-run versus Short-run

Producer Theory

13. Long-run vs Short-run

In the **long-run (LR)**, every input is variable

In the **short-run (SR)**, some inputs (not all of them) are fixed (usually capital)

There is no precise temporal definition, but the short-run represents a short period when we have limited flexibility in our actions.

On the other hand, long-run provides greater freedom.

13. Long-run vs Short-run

Thus, we can define C, AC, and MC functions for the short- and long-run:

$$C_{LR} = wl^* + rk^*$$

$$C_{SR} = rk_1 + wl$$

$$AC_{LR} = \frac{wl^* + rk^*}{q}$$

$$AC_{SR} = \frac{rk_1 + wl}{q}$$

$$MC_{LR} = \frac{\partial(wl^* + rk^*)}{\partial q}$$

$$MC_{SR} = \frac{\partial(rk_1 + wl)}{\partial q} = \frac{\partial(wl)}{\partial q}$$

13. Long-run vs Short-run

Variable Cost versus Average Variable Cost

In the short-run, total cost will be made of the sum between a **fixed cost** and a **variable cost** → Average variable cost (AVC) is equal to that variable cost divided by output (and MC also intersects it at its minimum)

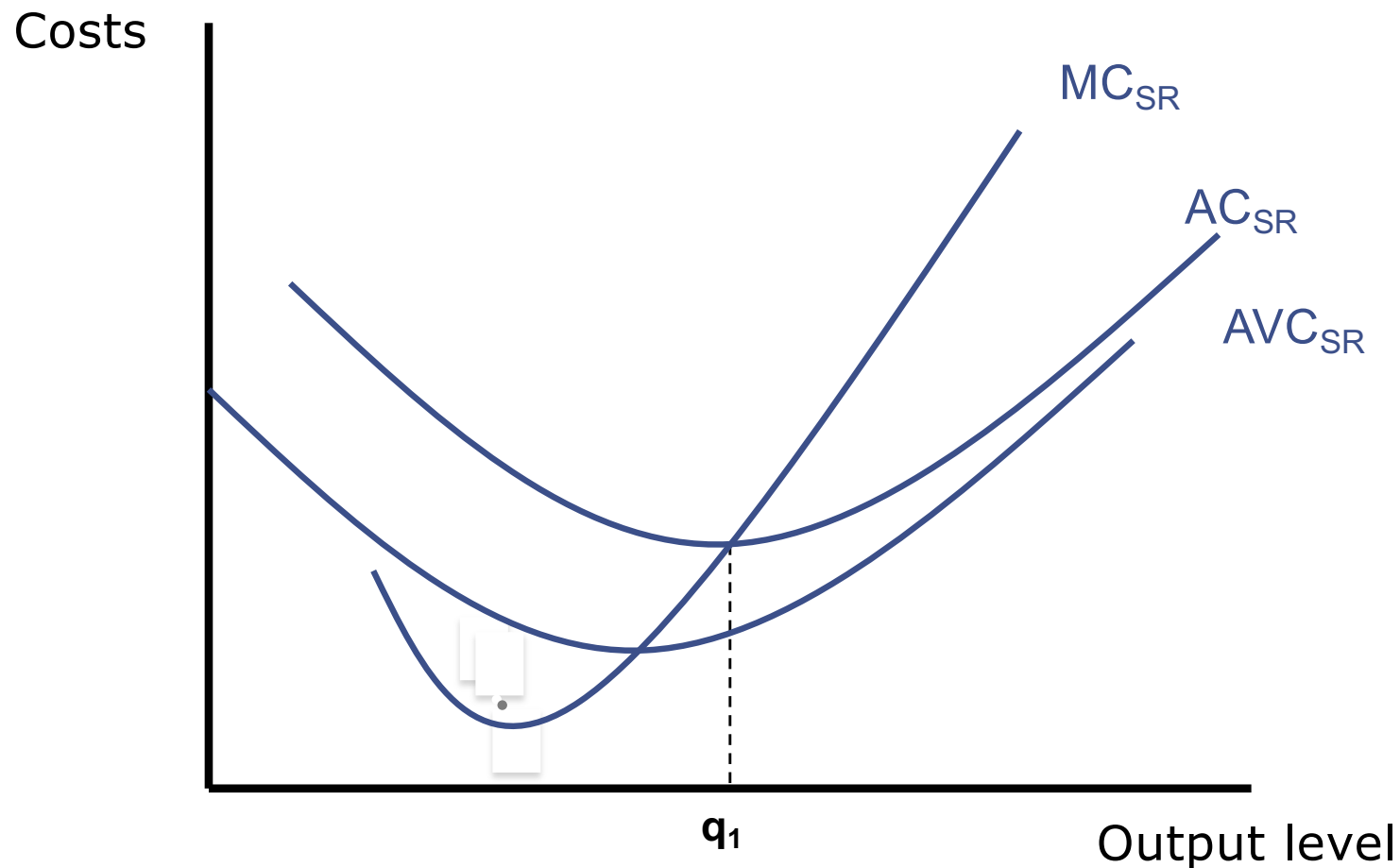
$$C_{SR} = \text{Fixed Cost} + \text{Variable Cost}$$

$$AC_{SR} = \frac{\text{Fixed Cost} + \text{Variable Cost}}{q}$$

$$AVC_{SR} = \frac{\text{Variable Cost}}{q}$$

13. Long-run vs Short-run

Variable Cost versus Average Variable Cost



Activity 2

A firm has the following production function $q = k^{1/3}l^{1/3}m^{1/3}$, where k is capital, l is labour, and m is materials. The prices are r , w , and t , respectively.

In the short-run, k and m are fixed.

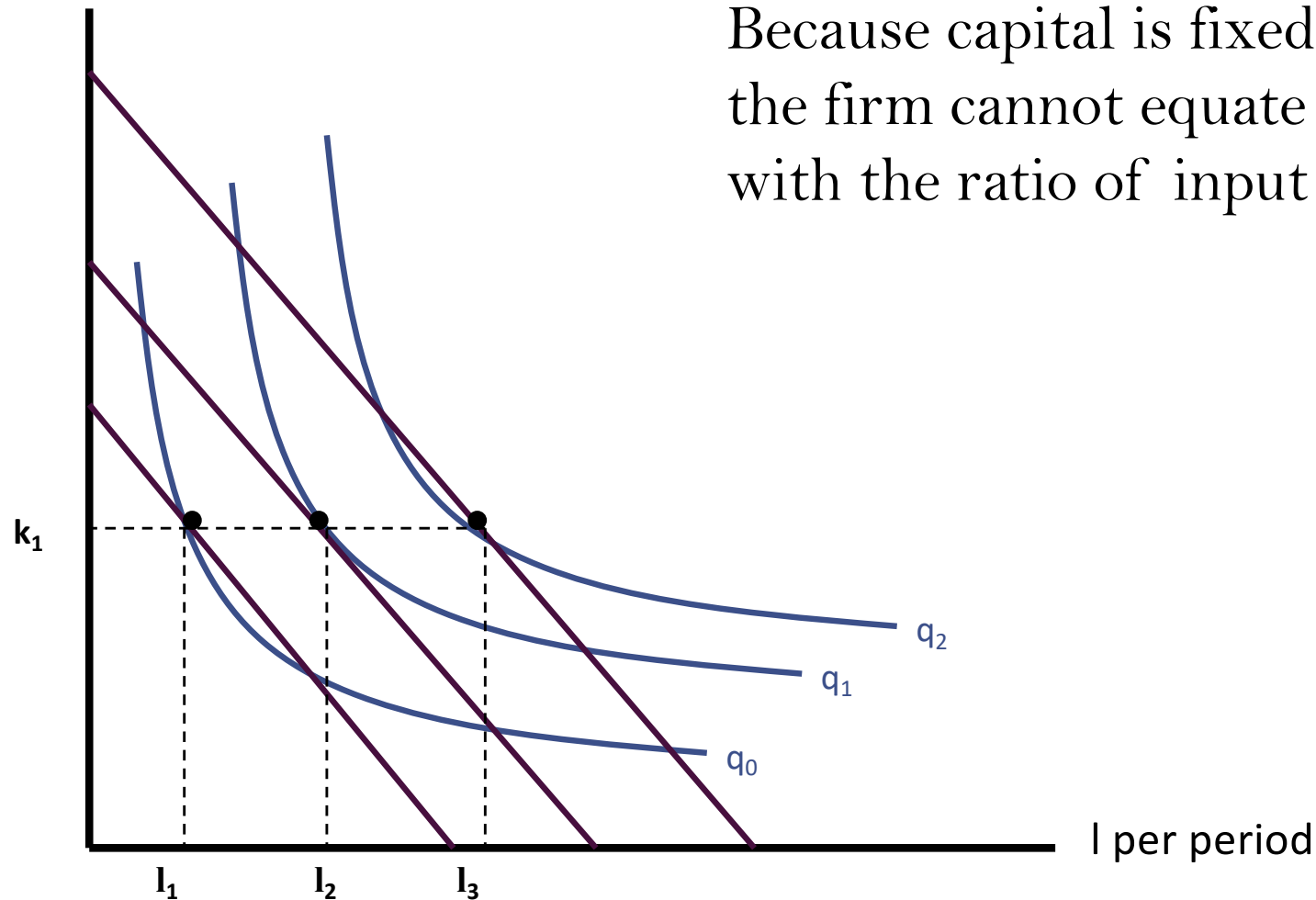
Find:

- a. Total Cost, Variable Cost, Average Cost, Average Variable Cost, and Marginal Cost in the **short-run**
- b. Total Cost, Average Cost, and Marginal Cost in the **long-run**

13. Long-run vs Short-run

Nonoptimality of the short-run costs

k per period



13. Long-run vs Short-run

The total cost curve and average cost curve in the Long-Run will be the **envelope** of the curves in the Short-Run

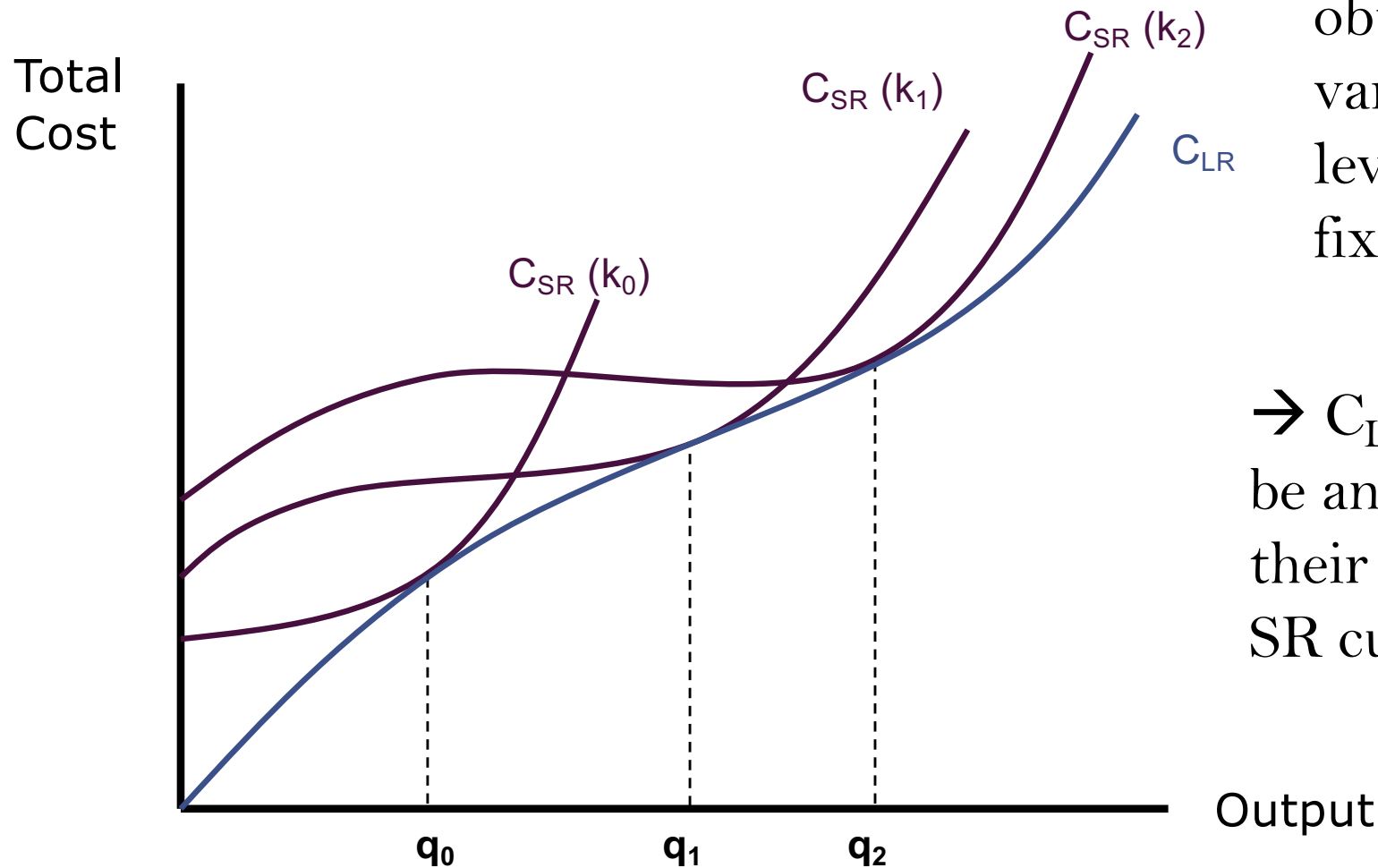
$$C_{SR} = C_{SR}(r, w, q, k_1)$$

The long-run total cost curve must obey the short-run relationship + the condition that k must be cost minimizing for any output level. A first-order condition for this minimization is that:

$$\frac{\partial C_{SR}(r, w, q, k_1)}{\partial k_1} = 0$$

13. Long-run vs Short-run

Relationships between Total Cost curves in the SR and LR

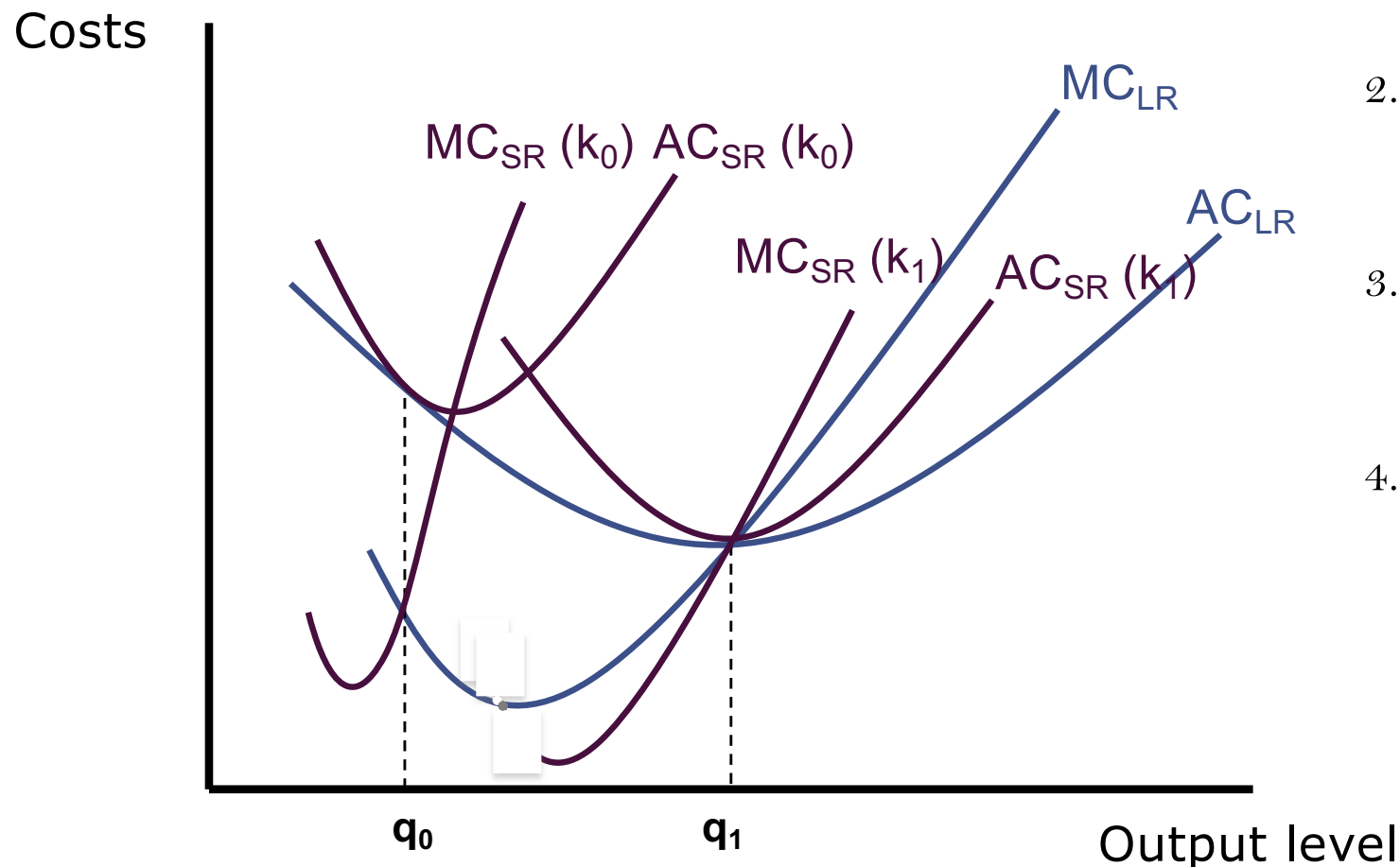


C_{LR} could be obtained by varying k 's level (which is fixed in the SR)

→ C_{LR} are said to be an **envelope** of their respective SR curves

13. Long-run vs Short-run

Relationships between AC and MC in the SR and LR



1. Each MC curve must intersect their AC curve at its minimum
2. AC_{LR} is the envelope of the AC_{SR} curves
3. MC_{LR} **is not** the envelope of the MC_{SR} curves
4. When the AC_{SR} intersects the AC_{LR} , at that level of production the MC_{SR} should intersect the MC_{LR}

Activity 2: Solution

- a. Total Cost, Variable Cost, Average Cost, Average Variable Cost, and Marginal Cost in the **short-run**

In the short-run, k and m are fixed $\rightarrow k_1$ and m_1

$$\text{Min } rk_1 + wl + tm_1$$

S. to:

$$k_1^{1/3} l^{1/3} m_1^{1/3} = q_0$$

Since k_1 and m_1 are given: $l_{SR} = \frac{q_0^3}{k_1 m_1}$

Activity 2: Solution

Thus,

$$C_{SR} = rk_1 + w\left(\frac{q_0^3}{k_1m_1}\right) + tm_1$$

$$C_{SR} = rk_1 + tm_1 + w\left(\frac{q_0^3}{k_1m_1}\right)$$

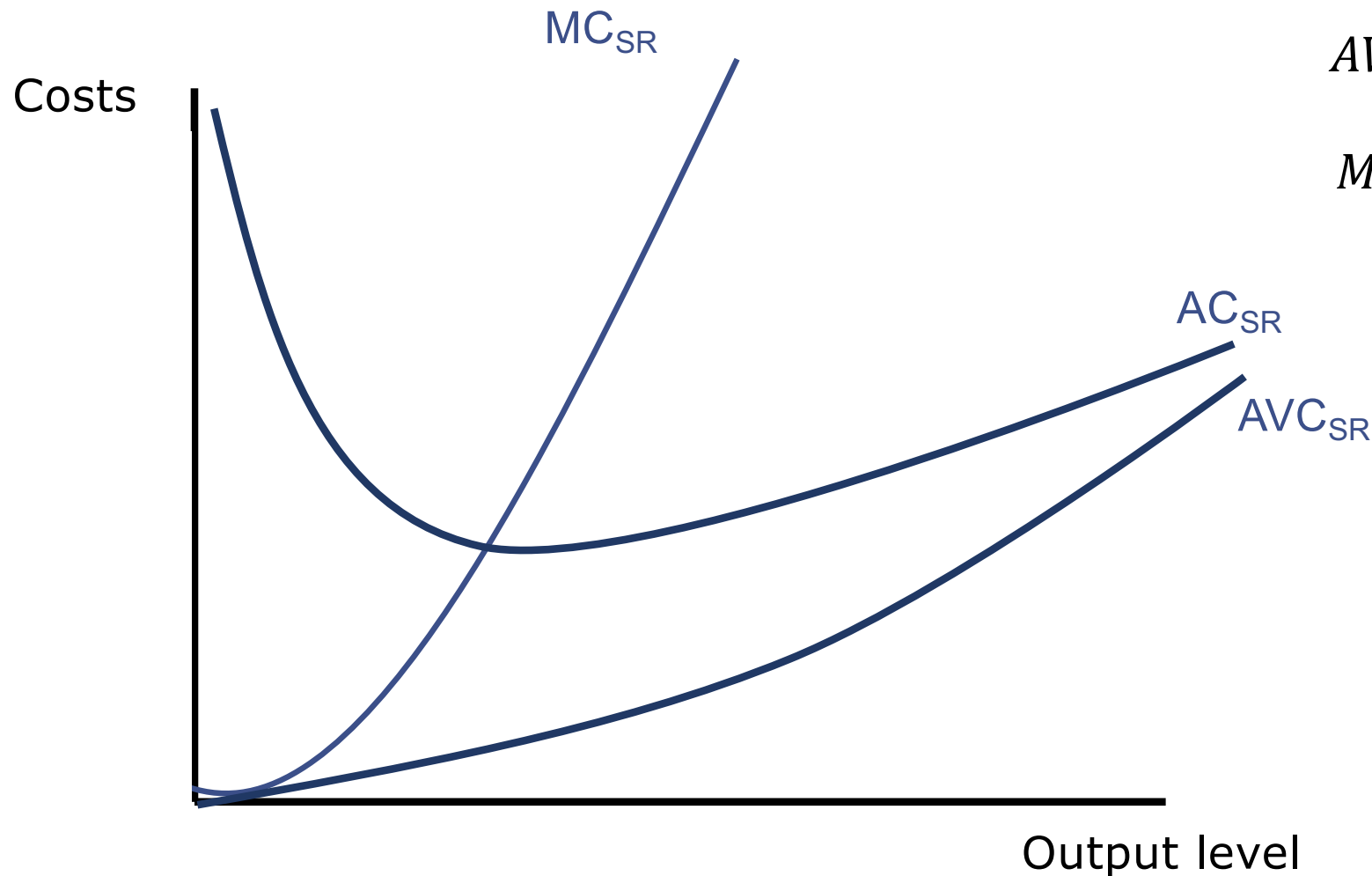
FIXED VARIABLE
COST COST

$$AC_{SR} = \frac{rk_1 + tm_1}{q_0} + w\left(\frac{q_0^2}{k_1m_1}\right)$$

$$AVC_{SR} = w\left(\frac{q_0^2}{k_1m_1}\right)$$

$$MC_{SR} = 3w\left(\frac{q_0}{k_1m_1}\right)$$

Activity 2: Solution



$$AC_{SR} = \frac{rk_1 + tm_1}{q_0} + w\left(\frac{q_0^2}{k_1 m_1}\right)$$

$$AVC_{SR} = w\left(\frac{q_0^2}{k_1 m_1}\right)$$

$$MC_{SR} = 3w\left(\frac{q_0^2}{k_1 m_1}\right)$$

Activity 2: Solution

- b. Total Cost, Average Cost, and Marginal Cost in the **long-run**

$$\text{Min } rk + wl + tm$$

S. to:

$$k^{1/3}l^{1/3}m^{1/3} = q_0$$

$$\mathcal{L} = r \cdot k + w \cdot l + t \cdot m + \lambda(q_0 - k^{1/3}l^{1/3}m^{1/3})$$

Activity 2: Solution

$$\mathcal{L} = r \cdot k + w \cdot l + t \cdot m + \lambda(q_0 - k^{1/3}l^{1/3}m^{1/3})$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial k} = r - \lambda \frac{1}{3} k^{-2/3} l^{1/3} m^{1/3} = 0$$

$$\frac{\partial \mathcal{L}}{\partial l} = w - \lambda \frac{1}{3} k^{1/3} l^{-2/3} m^{1/3} = 0$$

$$\frac{\partial \mathcal{L}}{\partial m} = t - \lambda \frac{1}{3} k^{1/3} l^{1/3} m^{-2/3} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q_0 - k^{1/3} l^{1/3} m^{1/3} = 0$$

Activity 2: Solution

$$\lambda = \frac{3r}{k^{-2/3} l^{1/3} m^{1/3}}$$

$$\lambda = \frac{3w}{k^{1/3} l^{-2/3} m^{1/3}}$$

$$\lambda = \frac{3l}{k^{1/3} l^{1/3} m^{-2/3}}$$

Thus,

$$\frac{3r}{k^{-2/3} l^{1/3} m^{1/3}} = \frac{3w}{k^{1/3} l^{-2/3} m^{1/3}}$$

$$\frac{r}{l} = \frac{w}{k} \rightarrow k = \frac{w}{r} l \quad \text{and, similarly, } m = \frac{w}{t} l$$

Activity 2: Solution

Replacing this equations in the restriction:

$$k^{1/3} l^{1/3} m^{1/3} = q_0$$
$$\left(\frac{w}{r} l\right)^{1/3} l^{1/3} \left(\frac{w}{t} l\right)^{1/3} = q_0$$

Thus,

$$l^* = q_0 \frac{(rt)^{1/3}}{w^{2/3}}$$
$$k^* = q_0 \frac{(tw)^{1/3}}{r^{2/3}}$$
$$m^* = q_0 \frac{(wr)^{1/3}}{t^{2/3}}$$

Activity 2: Solution

Replacing this optimal values in the cost:

$$C_{LR} = rk^* + wl^* + tm^*$$

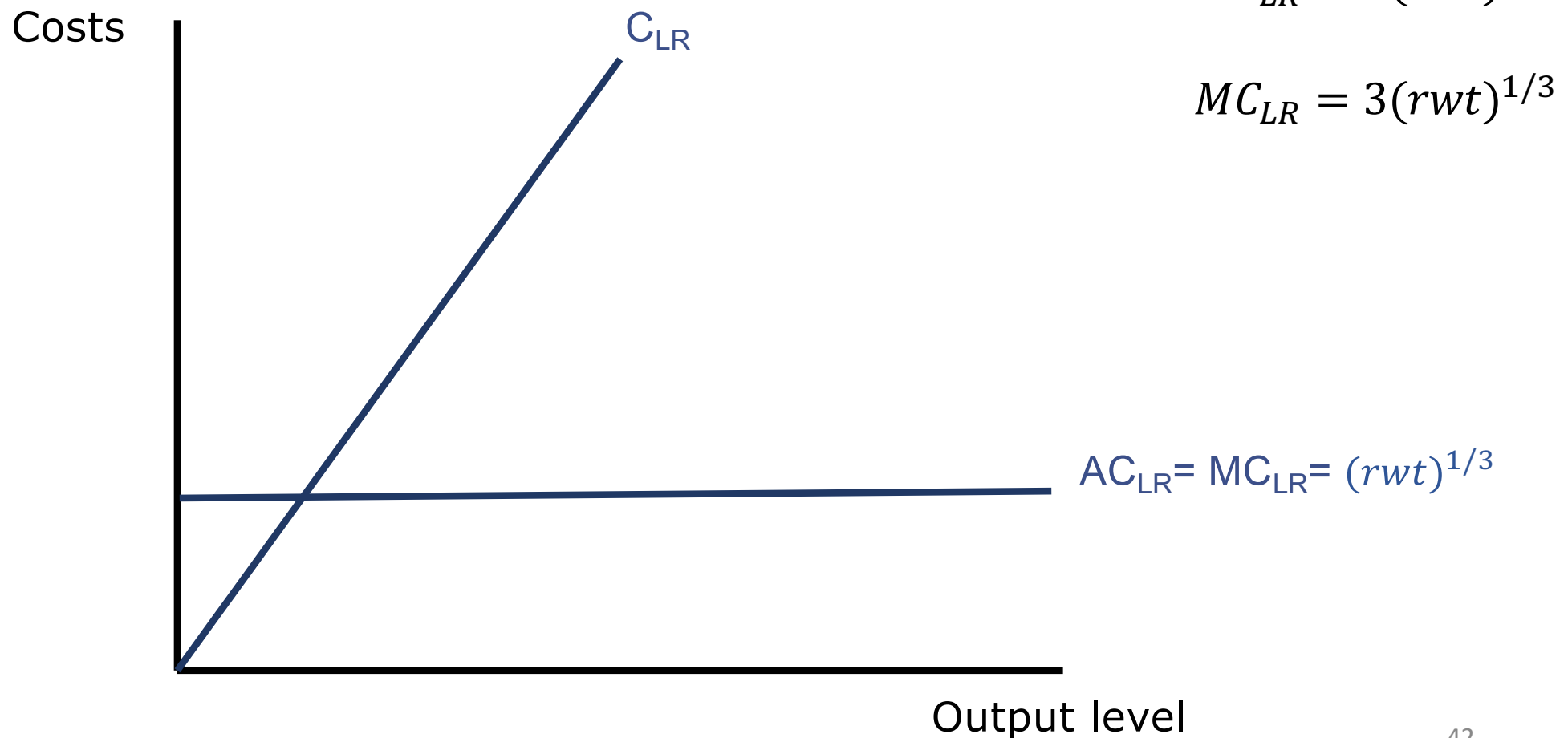
$$C_{LR} = r \cdot \left(q_0 \frac{(tw)^{\frac{1}{3}}}{r^{\frac{2}{3}}} \right) + w \cdot \left(q_0 \frac{(rt)^{\frac{1}{3}}}{w^{\frac{2}{3}}} \right) + t \left(q_0 \frac{(wr)^{\frac{1}{3}}}{t^{\frac{2}{3}}} \right)$$

$$C_{LR} = 3(rwt)^{1/3} q$$

$$AC_{LR} = 3(rwt)^{1/3}$$

$$MC_{LR} = 3(rwt)^{1/3}$$

Activity 2: Solution



Activity 3

A firm has the following production function $q = k^{1/3}l^{1/3}m^{1/3}$, where k is capital, l is labour, and m is materials. The prices are r , w , and t , respectively.

In the short-run, k and m are fixed.

Graph: Average Cost and Marginal Cost in the **short-run**
vs Average Cost and Marginal Cost in the **long-run**

Homework 1 Lecture 13

Imagine your firm faces a short-run cost function equal to:

$$C_{SR} = \bar{k}^2 + 16 \frac{Q^2}{\bar{k}}$$

- a. Find AC, MC, AVC in the short-run and graph them
- b. Find the Total Cost, AC, and MC curves in the long-run. Graph them.

HINT:

For b. remember that the total cost curve (or average cost curve) in the long run is the envelope of the total cost curve (or average cost curve) in the short run.

Homework 2 Lecture 13

A firm has the fixed proportion production function $q = \min(2l, 3k)$, where k is capital and l is labour. The prices are r and w , respectively.

In the short-run, k is fixed at 2.

Graph: Total Cost, Average Cost and Marginal Cost in the **short-run** vs Average Cost and Marginal Cost in the **long-run**

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

Activity 3: Solution

